Unit 1: Using Objects and Methods
Test Review

Name: _____

Multiple Choice

_____ 1) Consider the following code segment.

```
System.out.print("turtle ");
System.out.println("snake ");
System.out.println("dog ");
System.out.print("cat ");
```

What is printed as a result of executing the code segment?

(A) turtle
snake
dog cat

(B) turtle
snake
dog
cat

(C) turtle snake
dog
cat

(D) turtle snake
dog cat

_____ 2) Consider the following code segment.

```
int x = 3;
int y = /* missing expression */
x = 1 + 2 * y;
System.out.print(x);
System.out.println(y);
```

Which of the following can be used as a replacement for `/* missing expression */` so that the code segment prints 94?

(A) `3;`
(B) `x - 1;`
(C) `x + 1;`
(D) `1 - 2 * x;`

_____ 3) Consider the following code segment.

```
int x = 10;
int y = 20;
/* missing code */
System.out.print(top / bottom);
```

Which of the following replacements for `/* missing code */` will cause an `ArithmeticException` to occur?

I.
```
int top = x - y;
int bottom = y - x;
```

II.
```
int top = 2 * x;
int bottom = y - top;
```

III.
```
int top = x + y;
int bottom = 2 * top;
```

(A) I only
(B) II only
(C) I and II only
(D) II and III only

_____ 4) Which of the following expressions will evaluate to 1?

I. `2 / 5 % 3`
II. `2 / (5 % 3)`
III. `2 / 5 + 1`

(A) I only
(B) II only
(C) I and II only
(D) II and III only

_____ 5) Consider the following code.

```
int j = 12;
int k = 6;
j += 2;
k += j;
System.out.print(j + " " + k);
```

What is printed when the code segment executes?

(A) 14 20
(B) 14 18
(C) 8 12
(D) 12 6

_____ 6) Consider the following code segment.

```
int a = 10;
double b = 10.7;
double c = a + b;
int d = a + c;
System.out.println(c + " " + d);
```

(A) 20 20
(B) 20.0 30
(C) 20.7 31
(D) Nothing will be printed because of an error

_____ 7) Which of the following statements will **NOT** result in a compiler error?

I. `int x = (double) 4.5;`
II. `int x = (int) 4.5;`
III. `int x = 4.5;`

(A) I only
(B) II only
(C) III only
(D) I and II only

_____ 8) Given the following initializations,

```
double x = 5;
double y = 2;
```

What is the value of `m` after the following statement is executed?

```
int m = (int)(x + x / y - x * y - x / (10 * y));
```

(A) –1
(B) –0.75
(C) –2
(D) 0

_____ 9) What is the output of the following code?

```
int a = 1, b = 2, c = 3;
a += b + c;
b += a + c;
c += a + b;
System.out.println(a + " " + b + " " + c);
```

(A) 3 3 4
(B) 5 4 3
(C) 5 8 13
(D) 6 11 20

_____ 10) What output is produced by the following line of code?

```
int myNumber = 4;
System.out.println("myNumber/3 = 1");
System.out.println(myNumber/3);
```

(A) 1.333333333333 = 1
1.333333333333

(B) 1 = 1
1

(C) myNumber/3 = 1
1.333333333333

(D) myNumber/3 = 1
1

_____ 11) What is the output of the following code?

```
String s = "ONION";
String first = s.substring(1,5);
String other = first.substring(1,4);
String last = other.substring(0,2);
System.out.println(last);
```

(A) I
(B) IO
(C) ION
(D) ONI

_____ 12) Consider the following code segment.

```
String s = "goodbye";
System.out.println(s.substring(s.substring(s.length() - s.indexOf("d")).length()));
```

What is printed as a result of executing the code segment?

(A) bye
(B) 3
(C) dbye
(D) odbye

_____ 13) Consider the following declaration

```
Tool thing = new Tool();
```

Which of the following best describes the situation?

(A) `Tool` and `thing` are both classes
(B) `Tool` is an object of the `thing` class type
(C) `thing` is a primitive value of the `Tool` type
(D) `thing` is an instance of the `Tool` class type

_____ 14) Which of the following correctly gives random numbers between –10 and 10, inclusive, to the variable `n`?

(A) `int n = (int) (Math.random() * 11) - 20;`
(B) `int n = (int) (Math.random() * 21) - 10;`
(C) `int n = (int) (Math.random() * 10) - 21;`
(D) `int n = (int) (Math.random() * 20) - 10;`

_____ 15) Consider the following code.

```
String q = "power";
String r = "brown";
System.out.println(q.indexOf(r.substring(3, 4)));
```

What is the output?

(A) 2
(B) 1
(C) 3
(D) w

_____ 16) A student has created a `Book` class. The class contains variables to represent the following:

- An `int` variable called `pages` to represent the number of pages
- A `boolean` variable called `isHardcover` to indicate whether or not the book is hardcover
- The object `story` will be declared as type `Book`.

Which of the following descriptions is accurate?

(A) An instance of the `story` class is `Book`.
(B) An instance of the `Book` object is `story`.
(C) An attribute of the `story` object is `isHardcover`.
(D) An attribute of the `Book` instance is `story`.

_____ 17) Consider the following class declaration.

```
public class Thing {
     private String color;

     public Thing() {
          color = "Blue";
     }

     public Thing(String c) {
          color = c;
     }
}
```

Which of the following code segments, when appearing in a class other than `Thing`, will NOT compile correctly?

(A) `Thing someThing = new Thing("Green");`
(B) `Thing someThing = new Thing();`
(C) `Thing someThing;`
(D)
```
Thing someThing;
something = Thing();
```

_____ 18) Consider the following code segment.

```
String word = "September";
String str1 = word.substring(0, 3);
String str2 = word.substring(word.length() - 3);
System.out.println(str1 + str2);
```

What is printed when the code segment is executed?

(A) epber
(B) Sepbe
(C) Seper
(D) Sepber

_____ 19) Consider the following methods.

```
public void trial(){
      int a = 10;
      int b = 5;
      doublevalues(a,b);
      System.out.print(b);
      System.out.print(a);
}
public void doublevalues(int c, int d){
      c = c * 2;
      d = d * 2;
      System.out.print(c);
      System.out.print(d);
}
```

What is printed as the result of the call to `trial()`?

(A) 2010105
(B) 2010510
(C) 20102010
(D) 20101020

_____ 20) Consider the following class declaration.

```
public class VetRecord {
      private String name;
      private int age;
      private int weight;
      private boolean needsVaccine;

      public VetRecord(String nameP, int ageP, int weightP, boolean needsVaccineP) {
            name = nameP;
            age = ageP;
            weight = weightP;
            needsVaccine = needsVaccineP;
      }

      public VetRecord(String nameP, int ageP, int weightP) {
            name = nameP;
            age = ageP;
            weight = weightP;
            needsVaccine = true;

      }
}
```

A new constructor is to be added to the `VetRecord` class. Which of the following is NOT a possible header for the new constructor?

(A) `public VetRecord(int ageP, boolean needsVaccine)`
(B) `public VetRecord(String nameP, int ageP)`
(C) `public VetRecord(String nameP, boolean needsVaccine)`
(D) `public VetRecord(String nameP, int weightP, int ageP)`

Free Response

21) Give 4 code examples of how to increment the integer `j` by 1.

22)

a) For the following, give the range of possible values that can be displayed.	b) For the following, write a statement that generates a random number in the specified range (inclusive).
`Math.random() * 8`	0 to 10
`(int)(Math.random() * 20)`	1 to 500
`(int)(Math.random() * 20) + 1`	10 to 15
`(int)(Math.random() * 3 + 10)`	25 to 50
`(int)(Math.random() * 100) + 16`	–10 to 15

23) Consider the following initialization.

```
String name = "Godzilla";
```

What are the values of the following code segments?

`name.length()`	`name.substring(4)`	`name.substring(0,3)`
`name.substring(0,0)`	`name.indexOf("i")`	`name.indexOf("il")`
`name.indexOf("za")`	`name.compareTo("D")`	`name.compareTo("Gt")`

24) Write a class called `Sphere`. This class should have the following:

- one `double` instance variable that represents the radius
- a default constructor that sets the radius to 1
- a parameter constructor that sets the radius to a passed value
- a method that returns the surface area using the formula $4\pi r^2$, where r represents the radius of the sphere
- a method that returns the volume using the formula $\frac{4}{3}\pi r^3$, where r represents the radius of the sphere